

# Chemistry

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## Unit 2

### Practice Exam 2 Answers

#### Section one: Multiple-choice

##### Question 1

- D** As methanol has a higher vapour pressure than ethanol, more methanol molecules will be in the gas phase. The intermolecular bonds between methanol molecules are weaker and consequently the boiling point will be lower.

##### Question 2

- B** The oxygen atom has 6 electrons in the valence shell. Only 2 of the electrons are involved in bond formation in the  $O_2$  molecule, leaving 4 electrons or 2 lone electron pairs on each of the atoms in the  $O_2$  molecule. There is a total of 4 lone pairs of electrons in a molecule of oxygen.

##### Question 3

- B**  $O_2$  and  $N_2$  are non-polar as there is no electronegativity difference between atoms of the same element. There is a slight electronegativity difference between C and H atoms in  $CH_4$ . Overall the  $CH_4$  molecule is non-polar because of the tetrahedral symmetry of the molecule. While individual C-O bonds in  $CO_2$  are polar, the molecule is linear and the individual dipoles are cancelled out because of molecular symmetry.

##### Question 4

- C** The compound shown has two highly polar O-H bonds that are capable of forming hydrogen bonds with water. Compound A has no polar bonds and is insoluble in water. Compound B has one polar O-H bond and forms fewer hydrogen bonds with water than compound C. The intermolecular forces between water and compound D are dipole-dipole interactions which are not as strong as hydrogen bonds.

##### Question 5

- D** The formation of a solution involves breaking bonds between solute particles, breaking intermolecular bonds between solvent molecules, and the formation of bonds between solvent molecules and solute particles.

##### Question 6

- A** The concentration of  $H^+$  ions is greater in the HCl solution than in the  $CH_3COOH$  solution because HCl is a strong acid and completely ionised in solution and  $CH_3COOH$  is a weak acid and partly ionised. Hence the pH of HCl will be lower than that of  $CH_3COOH$ , although both solutions will have a pH below 7. Water is neutral and has a pH of 7. NaOH is a strong base and will have a pH greater than 7.

### Question 7

- C** Magnesium reacts with hydrochloric acid to produce hydrogen and a solution of magnesium chloride. Chloride ions are spectator ions and take no part in the reaction. Magnesium metal is converted to  $\text{Mg}^{2+}$  ions and  $\text{H}^+$  ions are converted to  $\text{H}_2$  gas. Alternative C shows the correct formula for the reactants and products and is balanced with respect to the number of atoms of each element and with respect to electric charge.

### Question 8

- A**  $C_1V_1 = C_2V_2$  where  $C_1 = 0.100 \text{ M}$ ,  $V_1 = 10.0 \text{ mL}$ ,  $C_2 = 0.04 \text{ M}$   
 $V_2 = 10.0 \times 0.100/0.04$   
 $= 25.0 \text{ mL}$

The original volume of acid was 10.0 mL and the final volume of the diluted acid was 25.0 mL. 15 mL of water must be added to dilute the acid to the required concentration.

### Question 9

- A** An ionic equation only include ions that are involved in the reaction.  $\text{Na}^+$  and  $\text{Cl}^-$  ions are not included in the ionic equation as they are spectator ions.  $\text{Ba}^{2+}$  ions combine with  $\text{SO}_4^{2-}$  ions to form barium sulfate, an insoluble salt that will precipitate out of solution.

### Question 10

- C** From the equation  $n(\text{CO}_2) = 3 n(\text{C}_3\text{H}_8)$   
 $= 3 \times 500/(3 \times 12.01 + 8 \times 1.008)$   
 $= 3 \times 500/44.09$   
 $= 34.02 \text{ mol}$   
 $V(\text{CO}_2) = 34.02 \times 22.71$   
 $= 773 \text{ L}$

### Question 11

- C** **A**  $m(\text{Co}) = 0.25 \times 58.93 = 14.73 \text{ g}$   
**B**  $14.0 \text{ g of } (\text{NH}_4)_2\text{SO}_4$   
**C**  $n(\text{CO}_2) = 2.4 \times 1023/6.02 \times 10^{23}$   
 $= 0.40$   
 $\Rightarrow m(\text{CO}_2) = 0.40 \times 44.01 = 17.5 \text{ g}$   
**D**  $n(\text{He}) = 98/22.71 = 4.32$   
 $\Rightarrow m(\text{He}) = 4.32 \times 4.003 = 17.3 \text{ g}$   
so **C** has the greatest mass.

### Question 12

- B** All of the other reactions involve production of a gas. If the reaction takes place in an open beaker the gas will escape and the mass of chemicals in the reaction vessel will decrease. No gas is produced in reaction B and the mass of chemicals in the reaction vessel will not change.

**Question 13**

- A** The rate of a reaction depends on concentration, temperature and surface area. In this example concentration is kept constant. The reaction is expected to be faster at 60°C than at 20°C. Magnesium powder has a larger surface area than the same mass of magnesium ribbon. The reaction involving magnesium powder added to acid at 60°C will be faster.

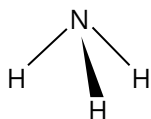
**End of section one**

## Section two: Short answer

### Question 14



b Triangular pyramid (1 mark)



c Nitrogen is a highly electronegative element and consequently the bonds between the atoms of nitrogen and hydrogen are highly polarised\*. The ammonia molecule is a dipole because of its shape and its polar bonds. There is an electrostatic attraction between the nitrogen atom of one ammonia molecule and the hydrogen atom of a nearby ammonia molecule\*. The attractions between these dipoles are known as hydrogen bonds\*. (3 marks)

### Question 15

- a Prepare a mixture of styrene and ethylbenzene. Pass this mixture through a gas chromatogram and note the retention time for each of the two peaks\*. Next, pass a sample of the reaction mixture through the same chromatogram and again note the retention times. If the number of peaks and their retention times are the same it can be assumed the reaction mixture contains ethylbenzene and styrene\*. (2 marks)
- b Prepare a series of styrene standards that have different concentrations of styrene. Pass these through the gas chromatograph measuring the area under the peak that is produced for each standard. Draw a calibration graph of peak area vs concentration\*. Pass the reaction mixture through the GC and measure the area of the peak produced by styrene. Read off the concentration from the calibration graph\*. (2 marks)

### Question 16

- a The volume of a fixed amount of gas at constant pressure is directly proportional to the temperature\* measured on the Kelvin or absolute scale\*. (2 marks)
- b A rise in temperature increases the average kinetic energy of the molecules in a gas\*. The molecules move faster and collide with the walls more often and with greater force. Consequently the volume of the gas increases if the pressure is kept constant\*. (2 marks)

### Question 17

- a A solution that cannot dissolve any more solute at the given temperature. (1 mark)
- b The three solutions can be identified by observing what happens when a small amount of solute is added to each solution. A small amount of solute added to an unsaturated solution will dissolve\*. A small amount of solute added to a saturated solution will not dissolve\*. A small amount of solute added to a supersaturated solution or even knocking the solution will cause the solute to crystallise out of the solution\*. (3 marks)
- c i  $5.67 \text{ g in } 250.0 \text{ mL} = 5.67 \times 1000/250 = 22.68 \text{ g per } 1000 \text{ mL}$   
 $= 22.7 \text{ g L}^{-1}$ \* (1 mark)
- ii  $n(\text{Na}_2\text{S}_2\text{O}_3) = 22.68/153.12 = 0.14343 \text{ mol}^*$   
 $c(\text{Na}_2\text{S}_2\text{O}_3) = 0.143 \text{ mol L}^{-1} \text{ (3 sig figs}^*)$  (2 marks)

iii  $C_1V_1 = C_2V_2$  where  $C_1 = 0.144 \text{ mol L}^{-1}$ ,  $V_1 = 250 \text{ mL}$ ,  $V_2 = 250 + 350 = 600 \text{ mL}$   
 $C_2 = C_1V_1/V_2 = 0.144 \times 250/600^*$   
 $= 0.0600 \text{ mol L}^{-1}$  (3 sig figs\*) (2 marks)

### Question 18

- a Solution A is 0.1 M ammonia solution. The conductivity evidence indicates that the solute is partially ionised in solution. The pH value indicates a weak base. The properties of this solution are consistent with those of an ammonia solution\*.

Solution B is 0.1 M glucose solution. The conductivity evidence indicates that the solute is not ionised in solution. The pH value indicates a neutral solution. The properties of this solution are consistent with those of a glucose solution\*.

Solution C is 0.1 M sodium hydroxide solution. The conductivity evidence indicates that the solute is completely ionised in solution. The pH value indicates a strong base. The properties of this solution are consistent with those of a dilute solution of sodium hydroxide\*.

Solution D is 0.1 M hydrochloric acid solution. The conductivity evidence indicates that the solute is completely ionised in solution. The pH value indicates a strong acid. The properties of this solution are consistent with those of a dilute solution of hydrochloric acid\*.

Solution E is 0.1 M ethanoic acid solution. The conductivity evidence indicates that the solute is partially ionised in solution. The pH value indicates a weak acid. The properties of this solution are consistent with those of a dilute solution of ethanoic acid\*. (5 marks)

- b i Prediction. The solid  $\text{Fe}(\text{OH})_2$  will dissolve in the acid.  
 $2\text{H}^+(\text{aq}) + \text{Fe}(\text{OH})_2 \rightarrow \text{Fe}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$  (3 marks)
- ii Prediction. The pungent odour of ammonia gas will be evident. Moistened red litmus will turn blue.  
 $\text{OH}^-(\text{aq}) + \text{NH}_4^+(\text{aq}) \rightarrow \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l})$  (3 marks)  
 (One mark for correct prediction, one mark for correct formula, one mark for correct coefficients)

### Question 19

- a  $n(\text{NaHCO}_3) = cV = 0.0500 \times 200/1000$   
 $= 0.0100 \text{ mol}^*$   
 $m(\text{NaHCO}_3) = nM$   
 $= 0.0100 \times (22.99 + 1.008 + 12.01 + 3 \times 16.00)^*$   
 $= 0.0100 \times 84.01$   
 $= 0.8401$   
 $= 0.840 \text{ g}$  (3 sig figs)\* (3 marks)
- b From the equation  
 $n(\text{HCl}) = n(\text{NaHCO}_3)$   
 $= 0.05 \times 23.0/1000$   
 $= 0.001150 \text{ mol}$   
 $c(\text{HCl}) = n/V$  ( $V = 35.0 \text{ mL} = 0.0350 \text{ L}$ )  
 $= 0.001150/0.0350$   
 $= 0.03285$   
 $= 0.0329 \text{ mol L}^{-1}$  \* (3 sig figs) (2 marks)

## Section three: Extended answer

### Question 20

- a**
- i** hydrogen bond (1 mark)
  - ii** covalent bond (1 mark)
- b** Hydrogen bonds are a relatively strong type of intermolecular bond\*. Hence a relatively large amount of energy is required to break the bonds between water molecules\* resulting in a high boiling point. (2 marks)
- c** As water freezes, water molecules slow down. In ice, each water molecule has formed hydrogen bonds to four other water molecules forming an open structure. In this solid structure the water molecules are further apart than in liquid water\*. Hence, ice is less dense than water and consequently it will float on water\*. (2 marks)
- d** Water is a polar molecule. It is able to form hydrogen bonds, dipole-dipole bonds\* and ion-dipole bonds\* with solute particles. (2 marks)

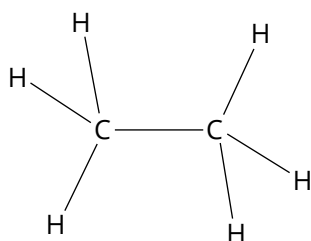
### Question 21

**a**

| element       | C     | H       |
|---------------|-------|---------|
| %             | 80    | 20      |
| atomic mass   | 12.01 | 1.008   |
| no. mole      | 6.66  | 19.84*  |
| mole ratio    | 1     | 2.98*   |
| simplest mole | 1     | 3 ratio |

The empirical formula of the compound is  $\text{CH}_3^*$ . (3 marks)

- b**  $n(\text{gas}) = V/22.71 = 5.15/22.71 = 0.227 \text{ mole}^*$   
molar mass =  $m/n = 6.72/0.227 = 29.60 \text{ g mol}^{-1*}$  (2 marks)
- c** Mass of empirical formula unit  $\text{CH}_3 = 12.01 + 3 \times 1.008 = 15.034^*$   
Ratio molar mass/empirical formula mass =  $28.54/15.03 = 1.90$   
Whole no. ratio =  $2^*$   
Molecular formula of the gas =  $2 \times \text{empirical formula} = \text{C}_2\text{H}_6^*$  (3 marks)
- d** Ethane\*  
(1 mark for diagram)



(2 marks)

- e** The intermolecular forces between hydrocarbon molecules are known as dispersion forces. They are the result of attraction between instantaneous dipoles that are formed by the random movement of electrons in molecules. (2 marks)

## Question 22

- a One mark each for correctly predicting formation of precipitate and one mark overall for predicting 'No change'. (4 marks)

|                    | Sodium carbonate                        | Sulfuric acid | Potassium nitrate | Magnesium nitrate | Copper(II) nitrate |
|--------------------|-----------------------------------------|---------------|-------------------|-------------------|--------------------|
| Sodium carbonate   |                                         |               |                   |                   |                    |
| Sulfuric acid      | A colourless, odourless gas is produced |               |                   |                   |                    |
| Potassium nitrate  | No change                               | No change     |                   |                   |                    |
| Magnesium nitrate  | A white precipitate is formed           | No change     | No change         |                   |                    |
| Copper(II) nitrate | A green precipitate is produced         | No change     | No change         | No change         |                    |

- b  $\text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})^*$   
 $\text{Cu}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{CuCO}_3(\text{s})^*$   
 $\text{Mg}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{MgCO}_3(\text{s})^*$  (3 marks)

- c Sodium carbonate solution will react with three of the other solutions. With one solution it will form a gas, with another solution it will form a white precipitate, and with the third it will form a green precipitate. No reaction will be observed with the fourth solution\*.

The sulfuric acid solution will react with only one other solution to produce gas bubbles\*.

Potassium nitrate solution will not react with any of the other four solutions\*.

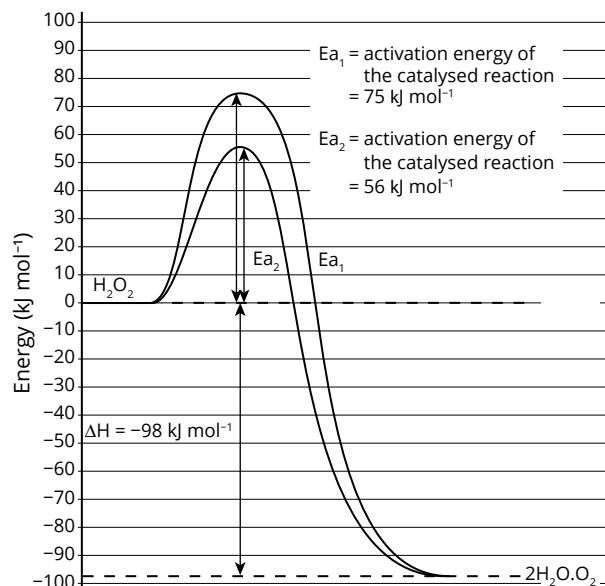
Magnesium nitrate solution will only react with one of the other solutions to form a white precipitate\*.

Copper nitrate solution will only react with one of the other solutions to form a green precipitate\*. (5 marks)

## Question 23

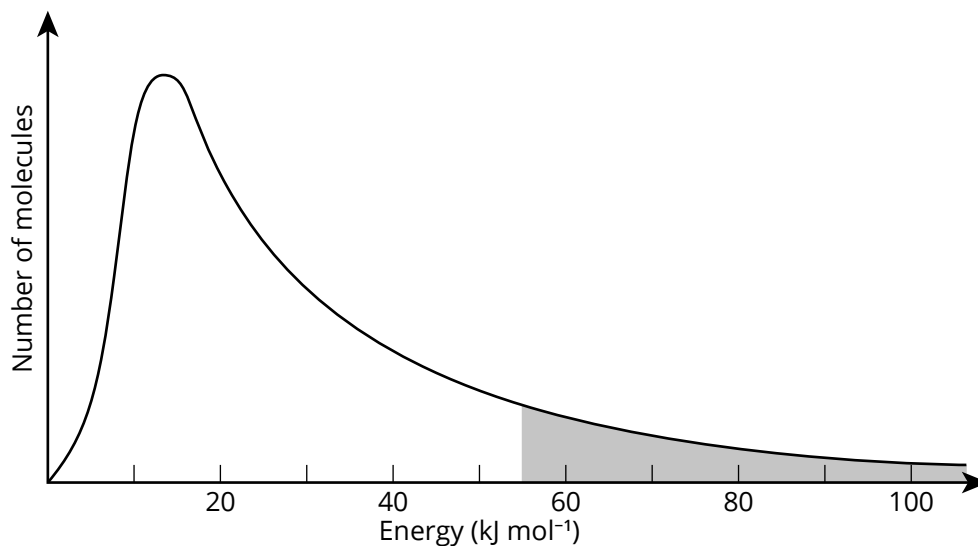
- a The activation energy is the minimum amount of energy required to break the bonds so an activation or transition complex can form enabling the reaction to proceed. (1 mark)

- b (3 marks)

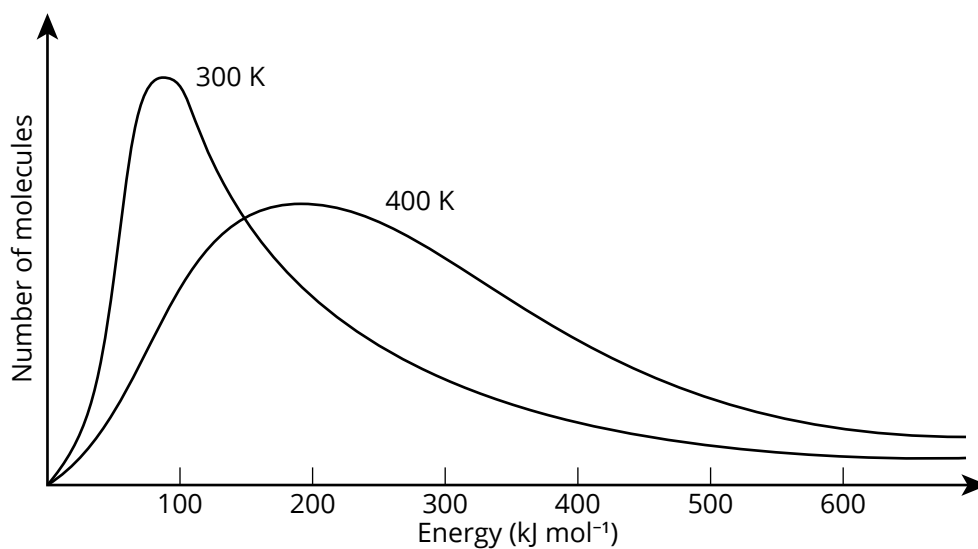


- c** The addition of an iodine catalyst enables the reaction to proceed along a different reaction pathway that has a lower activation energy, thus increasing the rate. (2 marks)

- d i** (1 mark)



- ii** (1 mark)



- iii** As the temperature is raised, particles move faster and their average kinetic energy increases\*. More particles have an energy equal to or greater than the activation energy\*. Consequently, a greater number of collisions will result in a reaction. (2 marks)

**End of answers**